

Technical Reference: Physics-Guided Calibrated Methane Localization

Complete training, statistical, and verification record — final 4–8-mast super-emitter benchmark

Branch `super-emitter`; data `csr-data-se48`

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1 Benchmark specification

Site and grid. 500×500 m site, 64×64 candidate grid (cell ≈ 7.8 m); sources snapped to cell centers, positions uniform on $[0.1, 0.9]^2$ (bitwise verified for all 15,000 scenarios).

Population. Release rate $q \sim \text{LogUniform}(100, 500) \text{ kg h}^{-1}$ (all super-emitters; realized range 100.0–500.0 in every split); masts $S \sim \text{Uniform}\{4, \dots, 8\}$ at uniform positions.

Transport. Steady-state Gaussian plume with ground reflection; Pasquill–Gifford stability (one class per scenario, uniform over A–F) with Briggs open-country dispersion (verified term-for-term against Hanna, Briggs & Hosker 1982); release height 3 m, sensor height 2 m; source-size floors $\sigma_{y0} = 2$ m, $\sigma_{z0} = 1$ m; wind-speed floor 0.5 m s^{-1} ; zero concentration upwind; $1 \text{ ppm CH}_4 = 6.55 \times 10^{-7} \text{ kg m}^{-3}$. Mass conservation verified to 4×10^{-6} relative.

Wind. Per scenario: mean speed $\text{LogUniform}(1, 8) \text{ m s}^{-1}$, uniform direction; per-step AR(1) meander ($\rho = 0.7$) with direction amplitude $\text{U}(20^\circ, 40^\circ)$ and log-speed amplitude $\text{U}(0.10, 0.30)$; $T = 30$ steps, each an instantaneous steady-plume snapshot.

Observation. $y = qg + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma \sim \text{LogUniform}(0.2, 3.0)$ ppm; each reading independently missing with probability $p_d \sim \text{U}(0, 0.2)$; all sums run over retained readings.

Anemometer error (the information gap). Recorded wind = truth corrupted per step: direction $+\mathcal{N}(0, (10^\circ)^2)$, speed $\times e^{\mathcal{N}(0, 0.10^2)}$. Empirical: 9.98° / 0.1000, no cross-step correlation. No model or map observes the true wind (verified bitwise, Sec. 8).

Splits. 10,000 train / 1,000 val (checkpoint selection) / 2,000 calibration (conformal threshold) / 2,000 test (reporting only). Zero content overlap on all six pairs. Seed streams (root 20260717): splits [root, 50..53, i]; measured wind [root, 90, tag, i]; oracle draws stream 91; map draws stream 92.

2 Physics input maps (3 channels, shared $K=8$ draws)

Per candidate cell c and wind draw $w^{(k)} \sim \mathcal{E}_{10^\circ, 10\%}(\hat{w})$, $k = 1..8$ (stream 92), sufficient statistics $a_c^{(k)} = g_c^\top y$, $b_c^{(k)} = \|g_c\|^2$, and $z_c^{(k)} = a_c^{(k)} / (\sigma \sqrt{b_c^{(k)}})$. Channels:

$$\begin{aligned} \text{evidence: } & \frac{1}{K} \sum_k \text{asinh}(z_c^{(k)} / 10) \\ \text{wind spread: } & 3 \times \text{std}_k \text{asinh}(z_c^{(k)} / 10) \\ \text{visibility: } & \frac{1}{K} \sum_k (\log_{10} b_c^{(k)} + 8) / 6 \end{aligned}$$

The asinh replaces a former hard clip at ± 60 that saturated $\approx 40\%$ of near-source 5×5 neighborhoods on this population (train-split measurement); asinh equals $z/10$ in the small-signal regime and compresses logarithmically (never flat) beyond. Ladder variants: zdet = evidence channel alone at $\hat{w}$ (1 ch); det = evidence + visibility at $\hat{w}$ (2 ch), both measured-wind, no draws.

3 Architectures

Shared: token = [Fourier_{1,2,4,8}(mast x), Fourier(mast y), Fourier(t), asinh(y/σ), $u_x(t)$, $u_y(t)$] (27-d); context MLP ($u, v, \log D, \log \sigma, S$) $\rightarrow 64 \rightarrow 64$; decoder $\rightarrow 1024 \rightarrow 4096$ logits. Aggregation differs:

Network	Aggregation	Parameters
DeepSets (Zaheer et al.)	shared MLP (256) + masked mean+max pool	4,870,080
Graph network (custom MPNN)	mast nodes, wind-aligned ordered edges, 2 rounds	5,610,048
Set Transformer (Lee et al.)	2×ISAB (4 heads, 16 inducing) + PMA(1)	6,528,448
Physics head (all)	Conv 3→32→32→1 (3×3), GELU, zero-init	10,433

Head joins at the logits: $\ell = f_\theta(x) + h_\phi(M(x))$; final conv weight and bias zero-initialized (verified: initial logits bitwise equal the base network’s).

4 Training protocol

AdamW, lr $3 \times 10^{-4} \rightarrow 3 \times 10^{-5}$ cosine ($T_{\max} = 200$), weight decay 10^{-4} , batch 256, 200 epochs, gradient-norm clip 1.0, bfloat16 autocast (validation in fp32), D4 symmetry augmentation (jointly on sensors, source, wind, and map channels via inverse-permutation gather), targets = one-hot smoothed by a Gaussian of 0.75 cells (truncated 3σ), checkpoint = best validation NLL. Seeds 1–3 (torch seed 6000 + $s + 1300$; numpy stream s). **Head warmup:** for map configurations, all head parameters have `requires_grad=False` during epochs 0–99 (AdamW acquires no state; verified the head stays exactly zero), joining from zero-init at epoch 100. Empirically, warmed runs reproduce their baseline’s validation trajectory to every printed digit through epoch 75 for DeepSets and GNN (all seeds, all ladder rungs); Set Transformer training is non-bit-reproducible run-to-run (CUDA attention backward under autocast), independent of maps/warmup. No per-architecture tuning of any kind.

Final validation NLL, seeds 1/2/3 — DeepSets base: 6.637, 6.681, 6.645; DeepSets+maps: 6.042, 6.090, 6.104; GNN base: 6.046, 6.022, 6.048; GNN+maps: 5.800, 5.758, 5.772; ST base: 6.131, 6.005, 5.989; ST+maps: 5.721, 5.737, 5.706; zdet: 6.209, 6.282, 6.260; det: 6.139, 6.203, 6.182.

5 Conformal procedure

Score (calibration scenario j): randomized cumulative mass $s_j = \sum_{c: P_j(c) > P_j(c^*)} P_j(c) + U_j \sum_{c: P_j(c) = P_j(c^*)} P_j(c)$, computed in the numerically exact tail form; threshold = the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score ($n=2000$, $\alpha = 0.1 \Rightarrow k = 1801$); region = smallest set of highest-probability cells reaching the threshold mass (deterministic, boundary-inclusive). Tie randomization uses a fresh `default_rng(777)` per calibration. Known asymmetry: the randomized score vs. deterministic region rule over-covers only for near-atomic maps (material for the exact reference, immaterial for the diffuse learned maps; all empirical coverages 0.887–0.916). Radius convention: $r = \sqrt{\text{cells}/4096/\pi} \times 500$ m.

6 Complete results

6.1 Per-run values (measured wind, test split, $\alpha = 0.1$)

Config	Coverage			Median radius (m)			<50 m (%)		
	s1	s2	s3	s1	s2	s3	s1	s2	s3
Physics-only ($K=8$)		1.000		282.1 (full site)			0.0		
DeepSets base	.914	.887	.899	99.9	97.3	99.6	0.05	0.0	0.0
DeepSets + maps	.916	.892	.891	75.3	73.5	73.6	16.3	17.3	17.1
GNN base	.904	.902	.896	67.6	69.3	69.1	6.9	1.9	1.3
GNN + maps	.903	.912	.890	62.3	61.9	61.1	27.4	27.2	28.6
ST base	.895	.891	.902	75.1	74.0	72.3	8.4	10.3	12.1
ST + maps	.898	.888	.906	64.3	63.9	65.2	29.3	29.6	28.2

Worst-seed Clopper–Pearson 95% coverage intervals all contain 0.90. Exact-model reference (known wind): median 4.4 m, coverage 0.958, 75.5% <50 m.

6.2 Paired effects (seed 1, 2,000 paired scenarios, 10^4 bootstrap)

	Δ Radius (m)	Δ (< 50 m) (pp)	smaller region
DeepSets	−24.6 [−25.8, −23.0]	+16.2 [+14.6, +17.9]	95.9%
GNN	−5.2 [−6.4, −4.2]	+20.5 [+18.6, +22.3]	83.2%
Set Transformer	−10.7 [−12.1, −9.5]	+20.9 [+19.1, +22.7]	87.4%

6.3 Component ladder (DeepSets, 3 seeds)

Configuration	Radius (m)	<50 m
Input-only	97.3–99.9	0.0–0.1%
+ evidence (zdet, 1ch, $\hat{w}$)	75.2–78.4	10.9–11.7%
+ visibility (det, 2ch, $\hat{w}$)	74.0–75.6	12.4–13.6%
+ wind spread (ens, 3ch, $K=8$)	73.5–75.3	16.3–17.3%

Monotone per seed on both metrics; evidence is the dominant rung (−21.3 to −23.2 m per seed). The final rung also draw-averages the first two channels (not purely additive).

6.4 Sensor-density stratification (seed 1; median m / <50 m)

	4 masts	5	6	7	8
n per stratum	412	427	355	403	403
DeepSets base	122 / 0%	101 / 0%	97 / 0%	97 / 0%	92 / 0%
DeepSets + maps	93 / 8%	76 / 16%	74 / 18%	71 / 16%	67 / 24%
GNN base	94 / 2%	69 / 6%	63 / 6%	62 / 9%	60 / 13%
GNN + maps	87 / 14%	66 / 25%	58 / 28%	59 / 33%	55 / 38%
ST base	97 / 4%	75 / 8%	72 / 7%	72 / 10%	67 / 12%
ST + maps	81 / 17%	66 / 28%	63 / 30%	60 / 33%	58 / 38%

6.5 Operating points (seed 1, warmed checkpoints)

	90% level			80% level		
	cov	med (m)	<50 m	cov	med (m)	<50 m
DeepSets + maps	.916	75.3	16.3%	.802	62.8	28.3%
GNN + maps	.903	62.3	27.4%	.809	52.2	45.8%
ST + maps	.898	64.3	29.2%	.792	52.5	45.6%

Both levels keep their stated guarantee; the 90% level is the headline, the 80% level a labeled triage operating point.

7 Statistical methodology

Ranges: min–max over three independent training seeds. Paired effects: per-scenario pairing on the identical 2,000 test scenarios; 10^4 bootstrap resamples of scenario indices; statistic = difference of medians (and of <50 m fractions); percentile intervals. Coverage uncertainty: exact Clopper–Pearson. Independent re-derivation with a different RNG reproduced every interval within Monte-Carlo error. Generating script: `scripts/se-paired.ci.py`.

8 Verification record (adversarial audits)

- Split disjointness: content hash, 0 overlaps on all 6 pairs; 15,000 unique ids.
- Leakage: stored maps recomputed from raw readings at the measured wind match *bitwise* (max diff 0.0); recomputation at the true wind differs grossly (oracle total variation 1.0). Measured winds regenerate bitwise from seeds.
- Oracle prior: integration over $[100, 500]$ verified numerically (recompute at $[10, 500]$ visibly differs; at $[100, 500]$, $TV \leq 7 \times 10^{-16}$).
- Warmup fairness: warmed map runs reproduce baseline training trajectories to printed precision through epoch 75 (12/15 runs; ST exceptions traced to CUDA attention nondeterminism, orthogonal to the comparison). AdamW verified to leave frozen zero-init head parameters untouched.
- Audit integrity: for all 24 audit files, mask popcounts equal stored sizes and coverage flags equal mask membership of the true cell (0/48,000 mismatches per file-set).
- Package: every number in `tables_se48.pdf` reproduced from raw arrays; both figures’ claims verified at 5 quantiles and every mast count.
- Self-calibration gate: conformal-on-exact coverage $0.958 \in [0.85, 0.985]$ (bound calibrated by a pre-launch review measurement).

9 Provenance, archives, and disclosures

Artifacts. Data: `/projectnb/rise-tower/eric1/csr-data-se48` (splits, measured winds, maps, oracles, 24 audit npzs with sizes + covered + packed masks, checkpoints). Results: `results/paired.wind.json`, `results/se48_table_deltas.ci.json`, `results_package/`. Decision log: `GATES.md` (seven dated entries for this benchmark).

Archived superseded configurations (all retained): full population (10–500 kg/h, 4–12 masts); 6–12-mast grid (16/18 runs; maps hurt strong baselines there — the density boundary); 2–8-mast partial grid; clipped evidence map era (`oldclip/`); joint-training era (`joint/`, including the seed-4 warmup pilot).

Disclosures for the paper. (1) Simulation-only; the maps use the generating forward model. (2) The population (4–8 masts, $q \geq 100$) and two method refinements (asinh evidence transform; head warmup) were selected adaptively during development; the warmup was validated by a prespecified pilot on a disjoint seed before adoption, and every before-state is archived. (3) On denser networks (6–12 masts) relational baselines close the gap without physics inputs. (4) Set Transformer training is not bit-reproducible (CUDA attention); all its results are

3-seed ranges. (5) The conformal tie rule over-covers only near-atomic maps (affects the exact reference, not the learned models).